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Smectic-like bundle formation of planktonic bacteria upon nutrient starvation†

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Bacteria aggregate through various intercellular interactions to build biofilms, but the effect of environmental changes on them remains largely unexplored. Here, by using an experimental device that overcomes past difficulties, we observed the collective response of *Escherichia coli* aggregates to dynamic changes in the growth conditions. We discovered that nutrient starvation caused bacterial cells to arrange themselves into bundle-shaped clusters, developing a structure akin to that of smectic liquid crystals. The degree of the smectic-like bundle order was evaluated by a deep learning approach. Our experiments suggest that both the depletion attraction by extracellular polymeric substances and the growth arrest are essential for the bundle formation. Since these effects of nutrient starvation at the single-cell level are common to many bacterial species, bundle formation might also be a common collective behavior that bacterial cells may exhibit under harsh environments.

1 Introduction

Under favorable environments, bacteria may proliferate by self-replication and eventually build densely organized communities, in particular biofilms.[1] Biofilms are widely distributed in various habitats, *e.g.*, oceans, soil, kitchens, and also we humans grow bacteria in our intestines to live in harmony with each other.[2] Biofilms are also involved in our daily lives and industries, helping to degrade organic waste and produce fine chemicals.[3] While biofilms benefit humanity in this way, they also cause a variety of problems in, *e.g.*, medical equipment[4] and industrial products.[5] Therefore, understanding the physiology of biofilms is a crucial mission across diverse disciplines.

During the typical biofilm formation process, bacteria aggregate through a variety of cell–cell interactions.[6] While cell adhesion to material surfaces is regarded as one of the most crucial events in biofilm formation,[7,8] cell–cell interactions mainly promote aggregation of planktonic cells that are not attached to the surface, which has also been shown to be important for biofilm construction.[9] In the case of *Escherichia coli*, the Ag43 antigen protein plays a major role in the aggregation process[6,10–12] *via* protein–protein interactions.[13] Extracellular polysaccharides and other debris also contribute through depletion attraction,[14–19] and diverse spatial patterns may emerge depending on how cells and these substances interact.[20–22] Type I pili are also known to be important in some cases.[12,23,24]

The expression state of extracellular molecules involved in cell–cell interactions, including Ag43, is generally highly dependent on the environment. Interestingly, it was reported that the relative importance of Ag43 and type I pili in aggregation depends on the environment: Ag43 has less effect on aggregation in Luria–Bertani (LB) broth than in minimal medium such as M9.[12] Besides, the secretion of proteins and polysaccharides, known as extracellular polymeric substances (EPS), is known to be sensitive to environmental changes such as nutrient starvation.[25–28]

Because of the sensitivity of cell–cell interactions to the environment, aggregates of planktonic cells may collectively respond to environmental changes. However, planktonic aggregates under time-dependent environments have been hardly investigated. A major difficulty is the necessity to change the environment in a controlled manner, maintaining spatial uniformity without adding hydrodynamic perturbations, which may influence the aggregates[29–31] and even destroy them in some cases.

Here, we overcome the difficulty by a membrane-type microfluidic device that we developed previously, namely the extensive microperfusion system (EMPS), which can uniformly change the environment surrounding bacterial populations without hydrodynamic perturbations.[32] With this device and using several strains of *E. coli*, we observed how planktonic aggregates respond to abrupt nutrient starvation. We discovered that, upon starvation, nematically ordered planktonic cells

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arranged themselves into bundle-shaped clusters, developing a structure akin to that of smectic liquid crystals. By employing a machine learning technique, we quantified the degree of this smectic-like bundle order. Moreover, we found that these bundles were destroyed if cell growth was resumed by switching back to the nutritious condition. We also found that inhibition of EPS secretion hampered the bundle formation significantly. These results suggest that both depletion attraction between cells, mediated by secreted EPS, and the arrest of cell elongation are involved in the bundle formation.

2 Results

2.1 Experimental setup and observations under steady growth conditions

Using EMPS (Fig. 1), we observed planktonic aggregates of non-motile *E. coli* (strain W3110[33]), formed in microfabricated wells on the substrate of the device. The absence of motility is important to prevent motility-induced dispersion of cells. The wells are circular and 11 μm deep, with diameters ϕ ranging from 110 μm to 330 μm (Fig. 1a), fabricated on a coverslip by the method described in ref. 32. Bacteria were confined in those wells by a porous membrane, attached to the substrate *via* biotin–streptavidin bonding. By supplying fresh medium through the porous membrane, we can control the culture condition in a spatially uniform manner, without hydrodynamic perturbations.[32] The device was placed in an incubation box maintained at 37 °C, and fresh medium was automatically supplied by a syringe pump (Fig. 1b).

We first observed the formation of planktonic aggregates under steady growth conditions (see Section 4.3). When the wild-type *E. coli* W3110 (denoted by WT; see Table 1 for the list of strains used in this work) was cultured in M9 minimum medium with glucose and amino acid [denoted by M9(glc. + a.a.); see Table 2 for the list of measurements], we observed planktonic aggregates, as well as chains of cells that apparently divided but remained connected (Fig. S1a and Movie S1, ESI†). Similar chains were also observed in earlier studies[34–36] and are known to be maintained by Ag43 localized on the pole of the cell.[34,36] Using the strain W3110 *flu*::*kan* (hereafter called the *flu*::*kan* mutant), for which the *flu* gene producing Ag43 was replaced with a kanamycin-resistant gene (*kan*), we confirmed that chain formation hardly appeared in this case and planktonic aggregates were mainly produced (Fig. S1b and Movie S2, ESI†). This indeed indicates the relevance of Ag43 to the chain formation.

We also observed that some cells adhered to the substrate in both the WT and *flu*::*kan* strains (see, *e.g.*, Movie S2, ESI†). To focus on planktonic aggregations without adhesion of cells, here we attempted to use a strain without flagella, which are known to be an important factor for adhesion of *E. coli* cells to surfaces.[37–39] Specifically, we used the mutant W3110 Δ*fliC* (hereafter called the Δ*fliC* mutant), for which the *fliC* gene producing flagella was deleted, and confirmed that adhesion of cells to the surface was significantly prevented (Fig. S1c, ESI†), while planktonic aggregates appeared to be almost identical to those of the WT and the *flu*::*kan* mutant (Movie S3, ESI†). We also found that chain formation was significantly suppressed in this case (Fig. S1c, ESI†). These results imply that flagella and Ag43 jointly contribute to the chain formation, in agreement with a recent report on similar chain formation observed for another strain of *E. coli*.[36] On the basis of these observations, in the following we will mainly use the Δ*fliC* mutant, to avoid both surface adhesion and chain formation. With this strain, we

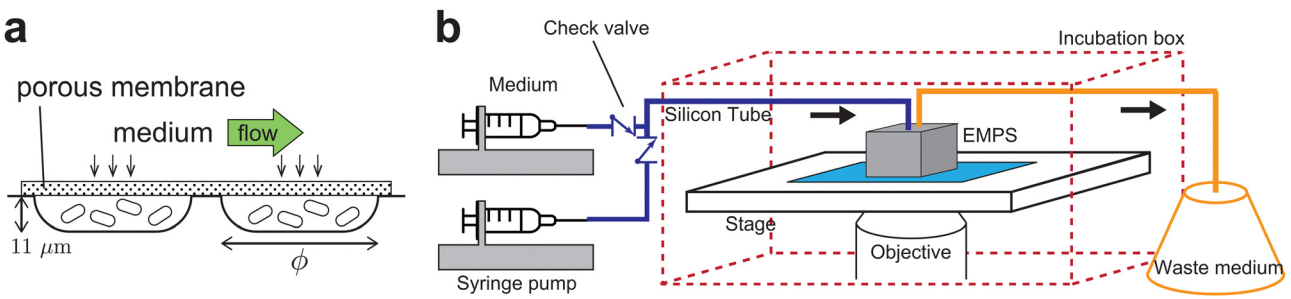


Fig. 1 Experimental setup. (a) A cross-sectional illustration of EMPS. Bacterial cells were captured in wells (depth 11 μm and diameter ϕ). The wells were covered by a rigid porous membrane, through which fresh medium was uniformly supplied. See ref. 32. (b) Overall setup of the experimental system. Liquid medium was constantly supplied from a syringe by a pump and drained into a waste container. By using another syringe with non-nutritious buffer, connected by a three-way junction with check valves, we can abruptly change the environment during observation.

Table 1 *E. coli* strains used in this study

Strains	Genotypes	Characteristics
WT	The wild-type of W3110	
flu::*kan* mutant	W3110 *flu*::*kan*	No Ag43
Δ*fliC* mutant	W3110 Δ*fliC*	No flagella
Δ*fliC csgA*::*kan* mutant	W3110 Δ*fliC csgA*::*kan*	No flagella or curli
Δ*fliC* (*venus*) mutant	W3110 Δ*fliC intS*::*venus-kan*	No flagella, expressing the fluorescent protein Venus

Table 2 List of measurements in this study. The column "Bundle fraction" indicates the fraction of the wells or dishes where the bundle formation was observed, showing the number of such wells as the numerator and the total number of the monitored wells as the denominator

Measurement	Strains	Conditions	Bundle fraction	Note
WT growth	WT	M9(glc + a.a.)		Fig. S1a and Movie S1 (ESI)
flu::*kan* growth	*flu*::*kan*	M9(glc + a.a.)		Fig. S1b and Movie S2 (ESI)
Δ*fliC* growth	Δ*fliC*	M9(glc + a.a.)		Fig. S1c and Movie S3 (ESI)
Δ*fliC* starvation	Δ*fliC*	M9(glc + a.a.) → PBS	30/30	Fig. 2a, b and Movie S4 (ESI)
Δ*fliC* (*venus*) starvation	Δ*fliC* (*venus*)	M9(glc + a.a.) → PBS		Fig. 2c and Fig. S5 (ESI), for fluorescence imaging
Δ*fliC* starvation (LB)	Δ*fliC*	LB → PBS	23/30	Fig. 2d and Movie S7 (ESI)
WT starvation	WT	M9(glc + a.a.) → PBS	15/18	Fig. 2e and Movie S8 (ESI), expressing flagella
Δ*fliC* starvation (BisBAL) #1	Δ*fliC*	M9(glc + a.a.) + 2 μM BisBAL → PBS + 2 μM BisBAL	14/30	Fig. 4 and Movies S9, S10 (ESI), reducing EPS production
Δ*fliC* starvation (BisBAL) #2	Δ*fliC*	M9(glc + a.a.) + 2 μM BisBAL → PBS + 2 μM BisBAL	4/13	Technical replicate
Δ*fliC csgA*::*kan* starvation	Δ*fliC csgA*::*kan*	M9(glc + a.a.) → PBS	15/19	Fig. S4 and Movie S11 (ESI), deleting curli
Δ*fliC* growth (PMB)	Δ*fliC*	M9(glc + a.a.) + 250 ng ml^{-1} polymyxin B	0/6	Fig. S6a and Movie S12 (ESI), promoting the production of OMVs
Δ*fliC* growth (LB, PMB)	Δ*fliC*	LB + 250 ng ml^{-1} polymyxin B	0/15	Fig. S6b and Movie S13 (ESI), promoting the production of OMVs
Δ*fliC* growth (xanthan)	Δ*fliC*	M9(Glc + a.a.) + 0.03 wt% xanthan gum	0/6	Fig. S6c and Movie S14 (ESI)
Δ*fliC* starvation-recovery	Δ*fliC*	M9(Glc + a.a.) → PBS → M9(Glc + a.a.)	25/27	Fig. 5 and Movie S15 (ESI)
Δ*fliC* starvation-dish (0.03% xanthan, BisBAL)	Δ*fliC*	PBS + 2 μM BisBAL + 0.03 wt% xanthan gum	3/3	Fig. S7 and Movies S16, S17 (ESI)
Δ*fliC* starvation-dish (0.015% xanthan, BisBAL)	Δ*fliC*	PBS + 2 mM BisBAL + 0.015 wt% xanthan gum	3/3	Fig. S7 (ESI)
Δ*fliC* starvation-dish (0.0075% xanthan, BisBAL)	Δ*fliC*	PBS + 2 μM BisBAL + 0.0075 wt% xanthan gum	2/3	Fig. S7 (ESI)
Δ*fliC* starvation-dish (0% xanthan, BisBAL)	Δ*fliC*	PBS + 2 μM BisBAL + 0 wt% xanthan gum	0/3	Fig. S7 (ESI)

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measured the time evolution of the total area of aggregates in microscopic images and verified that the growth rate of the aggregate size is comparable to the single-cell growth rate (Fig. S1d, ESI†). This indicates that the presence of aggregates does not deteriorate local growth conditions in the EMPS.

2.2 Bundle formation under nutrient starvation

We then observed how planktonic aggregates respond to nutrient starvation. We first grew bacteria (the Δ*fliC* mutant) in a nutritious condition to produce aggregates, then replaced the growth medium with non-nutritious phosphate-buffered saline (PBS) while observing the aggregates (see Section 4.3). Then we found, as shown in Fig. 2a, b and Movie S4 (ESI†), the formation of bundle clusters, where multiple cells clumped together next to each other. Such bundle clusters may also gather and develop a structure similar to that of the smectic A phase of liquid crystal (Fig. 2b). Bundle clusters can be observed more clearly by fluorescence microscopy (Fig. 2c) by

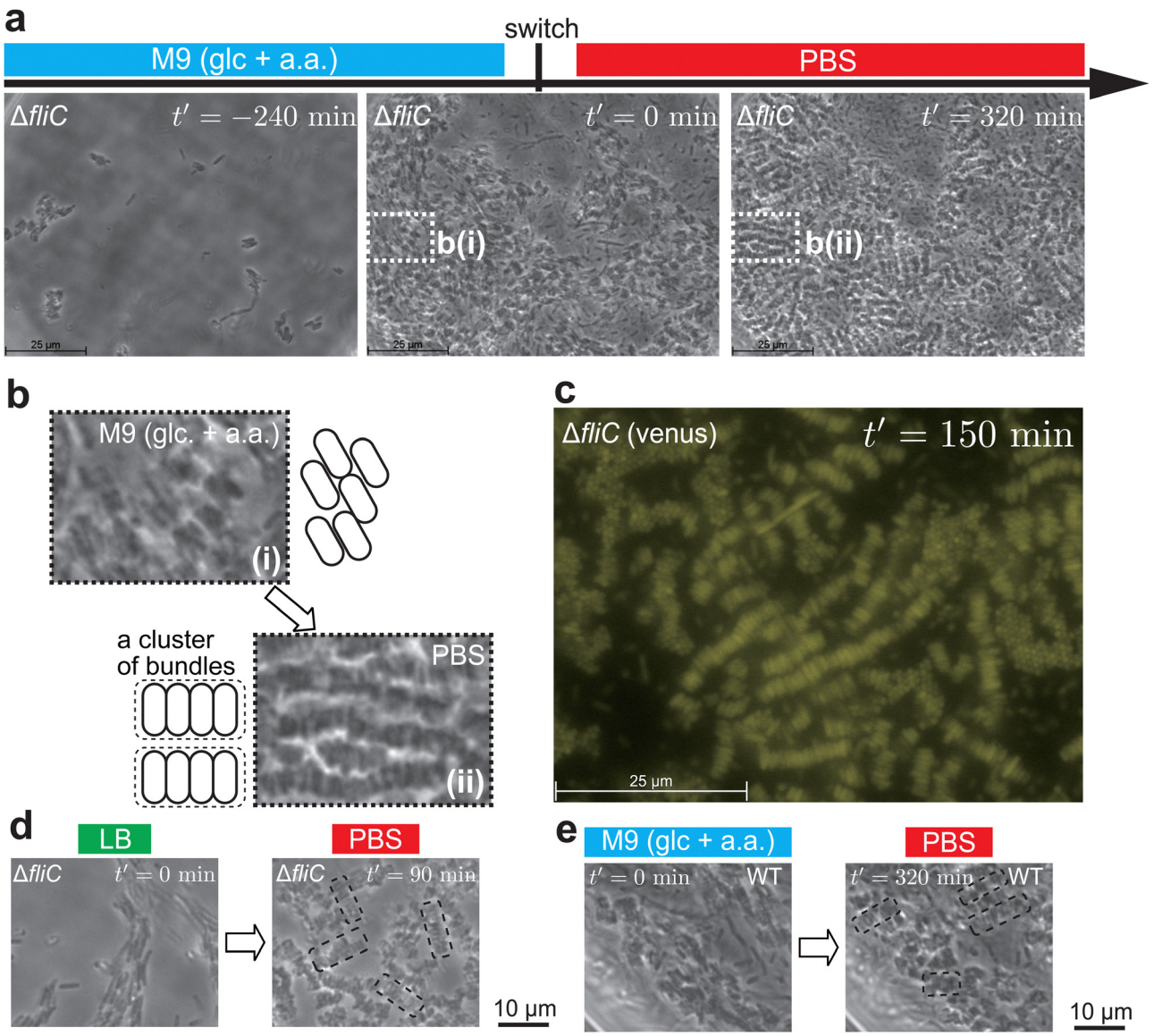


Fig. 2 Bundle formation upon nutrient starvation. (a) Observation of the Δ*fliC* mutant grown in M9(glc + a.a.) and starved in PBS. $t' = 0$ is the time at which the environment was switched. The well diameter ϕ is 150 μm. Bundle clusters were observed in all 30 wells with various diameters ϕ recorded in a single experiment ("Δ*fliC* starvation" in Table 2). See also Movie S4 (ESI†). (b) Enlargement of the regions outlined by the dotted rectangles in (a) at $t' = 0$ min (i) and $t' = 320$ min (ii). In (i), cells were nematically aligned. In (ii), these cells arranged themselves next to each other, and bundle clusters were formed. (c) A fluorescent image of bundles of the Δ*fliC* (*venus*) mutant grown in M9(glc + a.a.) and starved in PBS. The well diameter ϕ is 330 μm. (d) Observation of the Δ*fliC* mutant grown in LB and starved in PBS. The two snapshots were taken from the same position. Some regions with bundles are emphasized by dotted lines. The well diameter ϕ is 110 μm. Bundle clusters were observed in 23 out of 30 wells with various diameters recorded in a single experiment ("Δ*fliC* starvation (LB)" in Table 2). See also Movie S7 (ESI†). (e) Observation of the WT in the case of M9(glc + a.a.) → PBS. The two snapshots were taken from the same position. Some regions with bundles are emphasized by dotted lines. The well diameter ϕ is 150 μm. Bundle clusters were observed in 15 out of 18 wells with various diameters recorded in a single experiment ("WT starvation" in Table 2). See also Movie S8 (ESI†).

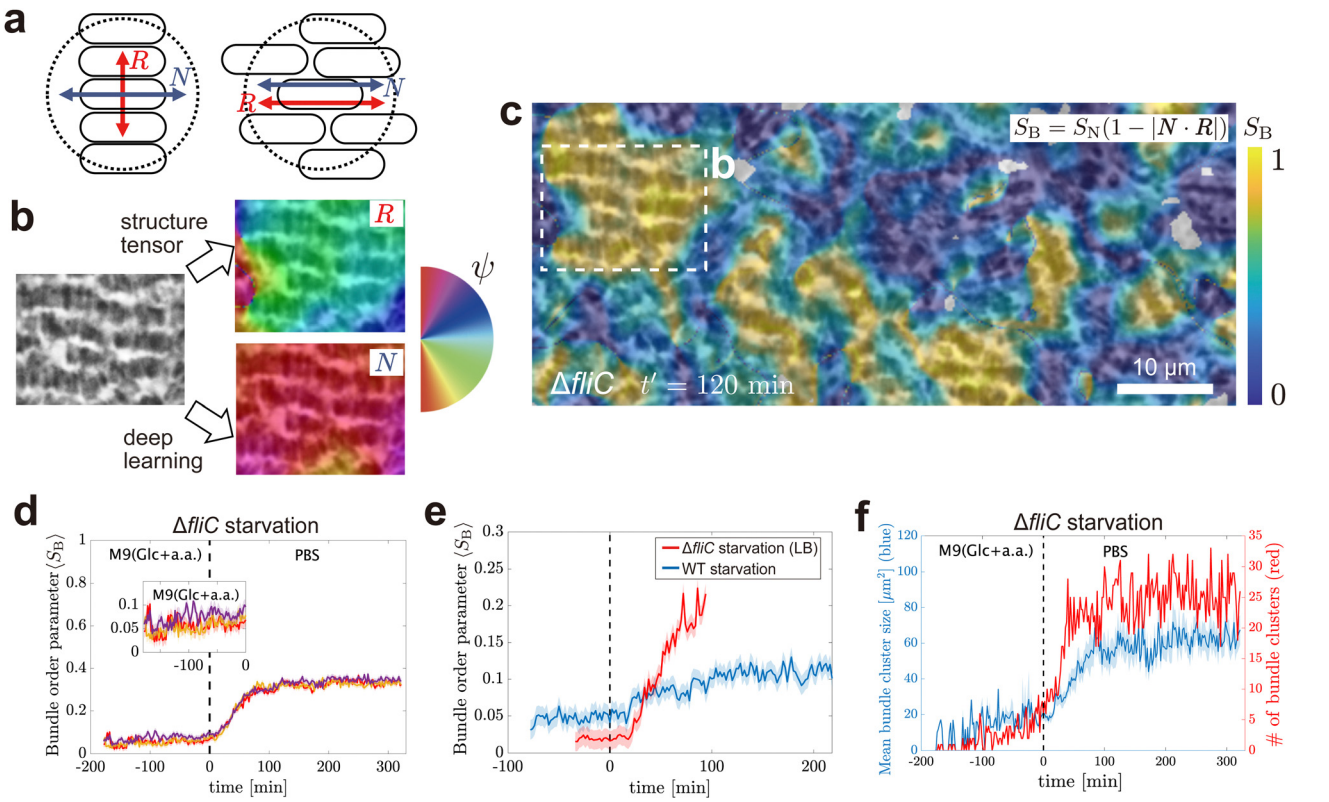


Fig. 3 Quantitative evaluation of the bundle formation. (a) Illustration of the definition of the density correlation director $\boldsymbol{R}$ and the cell orientation $\boldsymbol{N}$. (Left) When bundles are formed, $\boldsymbol{R}$ is oriented along the bundle axis, so that $\boldsymbol{R}$ and $\boldsymbol{N}$ are perpendicular. The dotted circle illustrates the approximate range of the Gaussian kernel used in eqn (7). (Right) Without bundles, $\boldsymbol{R}$ is along the long axis of cells, parallel to $\boldsymbol{N}$. (b) Estimation of $\boldsymbol{R}$ and $\boldsymbol{N}$. The image is an enlargement of the rectangular region in (c). The density correlation director $\boldsymbol{R}$ is obtained by the structure tensor method [eqn (7)]. The cell orientation is obtained by the deep learning model which we trained in this study. See Section 4.5 and Fig. S2 and S3 (ESI†). (c) The evaluated bundle order parameter field S_B. The image is from the Δ*fliC* starvation shown in Fig. 2a and b. (d) The spatially averaged bundle order parameter in the regions where aggregates exist, $\langle S_B \rangle$, for the Δ*fliC* starvation. Each time series represents the data in a separate well (ϕ = 150 μm). The inset shows an enlargement of the data before. The observation shown in (b), (c), Fig. 2a, b and Movies S4–S6 (ESI†) corresponds to the purple curve. The error is shown by the shaded bands, which was evaluated as the mean absolute error of the trained deep learning model compared to the ground truth by manual annotation (see Section 4.5 for details). (e) The spatially averaged bundle order parameter $\langle S_B \rangle$ for the Δ*fliC* starvation (LB) and the WT starvation. The former corresponds to the observation shown in Fig. 2d and Movie S7 (ESI†), and the latter to Fig. 2e and Movie S8 (ESI†). (f) The size and number of the bundle clusters for the Δ*fliC* starvation, measured in all three wells used in (d) at 20-minute intervals. The blue curve and the shaded band indicate the mean and standard error of the cluster sizes, respectively. The red curve represents the number of bundle clusters.

using a strain with a fluorescent protein, Venus.[40] We also observed that some cells were vertically aligned and formed a hexagonal lattice, suggesting that bundles may also be formed three-dimensionally. Moreover, the bundle formation was not limited to the specific condition and strain chosen here. We observed similar bundle clusters when we grew cells in LB medium (Fig. 2d and Movie S7, ESI;† "Δ*fliC* starvation (LB)" in Table 2) and when we used the WT expressing flagella (Fig. 2e and Movie S8, ESI;† "WT starvation" in Table 2). This indicates the robustness of our finding of the bundle formation upon starvation, regardless of detailed conditions and cellular states.

To quantify the degree of the smectic-like order, we define the bundle order parameter S_B by

$$S_B = S_N(1 - |\boldsymbol{N}\cdot\boldsymbol{R}|). \quad (1)$$

Here, $\boldsymbol{N}$ denotes the coarse-grained field of the cell orientation, $\boldsymbol{R}$ the orientation of the density correlation estimated from the image intensity correlation, and S_N the nematic scalar order parameter evaluated from the cell orientation $\boldsymbol{N}$ (Fig. 3a). The two orientations $\boldsymbol{R}$ and $\boldsymbol{N}$ were evaluated by the structure tensor method[41] and by a supervised deep neural network method, U-Net,[42] respectively (see Section 4.5 for details). The bundle order parameter S_B takes a large value when the nematic order is strong and $\boldsymbol{N}$ and $\boldsymbol{R}$ are nearly perpendicular. The latter condition is set because, when bundle clusters are formed, bacterial cells arrange themselves next to each other, in the direction of their short axis (Fig. 3b), similarly to the smectic A phase. We also observe that, when no bundles are formed, cells are typically located along their long axis, resulting in a stronger density correlation in that direction, $\boldsymbol{R}\|\boldsymbol{N}$. Thus, our bundle order parameter S_B can evaluate the degree of bundle formation and its smectic-like order, as demonstrated in Fig. 3c and Movies S5, S6 (ESI†). Although S_B cannot capture the bundle order of vertically aligned cells, such bundles typically take a minor fraction of area, so that S_B is able to characterize the degree of the bundle order of the population, at least

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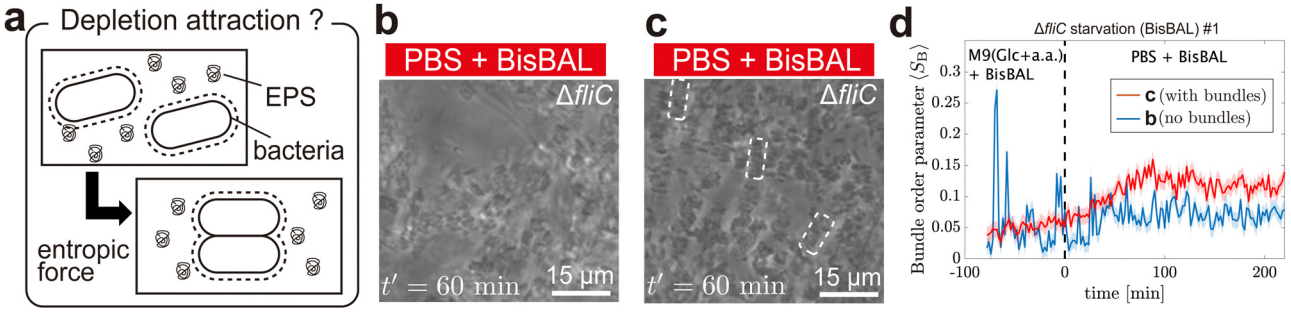


Fig. 4 Investigation of the relevance of EPS. (a) Illustration of the hypothesis that the depletion attraction is relevant to the bundle formation. The dotted lines around the bacterial cells represent the excluded region which large molecules such as EPS cannot enter. Cells may form bundles to minimize the excluded volume and increase the entropy of large molecules. (b) and (c) Two examples of starvation response of the Δ*fliC* mutant grown in M9(glc + a.a.) + 2 μM BisBAL and starved in PBS + 2 μM BisBAL. The well diameter ϕ is 150 μm. $t' = 0$ is the time at which the environment was switched. The bundle formation was not observed at all [(b) and Movie S9, ESI†] in 16 out of 30 wells with various diameters recorded in a single experiment (Δ*fliC* starvation (BisBAL) #1 in Table 2), while it was observed [(c) and Movie S10, ESI†] in the other wells. We also conducted a technical replicate (Δ*fliC* starvation (BisBAL) #2 in Table 2) and found no bundle formation in 9 out of 13 wells. These indicate that comprehensive reduction of EPS secretion by BisBAL significantly inhibits the bundle formation, suggesting the possible relevance of depletion attraction mediated by EPS. (d) The spatially averaged bundle order parameter $\langle S_B \rangle$ for the cases where bundles were formed and not formed.

qualitatively. Note also that, while S_B is different from the commonly used smectic order parameter,[43] it is more convenient when the coarse-grained density field is not readily available. A similar quantity was also adopted in the past[44] to evaluate the smectic order of colloidal rods.

To evaluate the bundle formation process, we first measured the time evolution of the spatially averaged bundle order parameter, $\langle S_B \rangle$. The spatial average was calculated over the regions where aggregates exist, which were also obtained by the deep neural network (see Section 4.5). We found that $\langle S_B \rangle$ increases significantly after the starvation, in all cases described so far [the Δ*fliC* starvation (Fig. 3d), the Δ*fliC* starvation (LB) and the WT starvation (Fig. 3e)].

In addition to the spatially averaged S_B, we also investigated the time evolution of the spatial structure. Specifically, we analyzed how the size and number of bundle clusters increased upon starvation. Here we defined a bundle cluster as a connected region that satisfies $S_B > 0.5$ with essentially uniform density orientation $\boldsymbol{R}$ (see Section 4.5 for a precise definition). We found that both the size and number of bundle clusters increase and converge during starvation (Fig. 3f). This shows that not only the spatially averaged order parameter but also the spatial structure of the bundles are developed and eventually stabilized.

We should note that the value of the bundle order parameter is sensitive to subtle differences in experimental conditions, such as the weakly heterogeneous image background due to the porous membrane. For example, in another measurement of the Δ*fliC* mutant strain from M9(glc. + a.a.) to PBS (specifically "Δ*fliC* starvation-recovery" in Table 2, to be described below), the bundle order parameter during the starvation (Fig. 5b) was significantly smaller than that of the Δ*fliC* starvation (Fig. 3d). Therefore, we compare the values of the bundle order parameter only within each technical replicate, to characterize the structure before and after starvation.

2.3 The relevance of extracellular molecules

Here we consider what changes in cell–cell interactions occur upon starvation and result in bundle formation. In the literature, qualitatively similar bundles of rod-shaped cells have been observed in the presence of a sufficient amount of polymer.[14,16,45–47] This was interpreted as a result of depletion attraction between cells, which results from the entropic force acting to minimize the excluded region around cells and lead to the lateral alignment of rod-shaped cells (Fig. 4a). A similar effect has also been reported for mixtures of rod-shaped and spherical colloids,[44,48–51] resulting in the smectic A phase similar to our observation. Since the secretion of some EPS is known to be promoted under external stresses,[25–28] we hypothesize that the EPS-induced depletion attraction may be the mechanism of the bundle formation upon starvation.

To test this hypothesis, we observed the starvation response of aggregates in the presence of a reagent that comprehensively inhibits EPS secretion. Specifically, we used bismuth dimercaprol (BisBAL), which has been reported to comprehensively reduce EPS secretion roughly by a factor of 10 under appropriate concentrations, without significantly affecting cell growth.[52,53] We cultured, in the presence of BisBAL, populations of the Δ*fliC* mutant forming aggregates in M9(Glc + a.a.) and exposed them to starvation in PBS ("Δ*fliC* starvation (BisBAL)" in Table 2). In contrast to the case without BisBAL, where bundles were formed in all monitored wells (Δ*fliC* starvation), with BisBAL we observed that bundles were not formed (Fig. 4b and Movie S9, ESI†) in more than half of the observed wells in two technical replicates (#1 and #2 in Table 2), while in the other wells bundles were formed relatively weakly (Fig. 4c and Movie S10, ESI†); see also the bundle order parameter for those two cases (Fig. 4d). This indicates that the reduction of EPS secretion by BisBAL significantly, albeit not completely, inhibits the bundle formation. Our results suggest the relevance of depletion attraction by EPS.

While the comprehensive inhibition of EPS secretion by BisBAL allowed us to know that EPS is involved in the bundle formation process, it is useful to identify specific substances that serve as depletants of this depletion attraction. There are several candidate substances, whose secretion is known to

increase during starvation and whose size is large enough to cause depletion attraction between cells. First, *E. coli* is known to promote the secretion of cellulose under environmental stresses.[54,55] However, we can immediately rule out its relevance because the W3110 strain used in this study does not produce cellulose.[55] Second, curli, polymeric amyloid fibers, which usually extend from and attach to cell surfaces,[56] are known to be produced more actively under environmental stresses.[56,57] It was reported that their major subunit protein, CsgA, can polymerize in medium being detached from cell surfaces,[58] which may induce depletion attraction. Third, outer membrane vesicles (OMVs) have a diameter of 20–250 nm, which contain, *e.g.*, misfolded proteins and toxic substances, and are discharged from cells.[59] It has been reported that the secretion of OMVs is promoted under external stress[59,60] and that they can induce depletion attraction to particles on the scale of tens of micrometers.[15,18] Therefore, we tested the possible influence of curli and OMVs on the bundle formation.

We first tested the possibility of curli, by using the mutant W3110 Δ*fliC csgA*::*kan* (''Δ*fliC csgA*::*kan*'' mutant in Table 1), for which the gene expressing CsgA protein was replaced by a kanamycin-resistant gene, *kan*. On an agar plate of LB broth containing Congo red, which selectively stains amyloid fibers including curli,[61] we confirmed that the Δ*fliC csgA*::*kan* mutant does not produce curli (Fig. S4a, ESI†). We then observed aggregates of the mutant grown in M9 (glc + a.a.) and starved in PBS (''Δ*fliC csgA*::*kan* starvation'' in Table 2) and found bundle clusters in most wells (Table 2 and Fig. S4b, Movie S11, ESI†). These indicate that curli may not be the main depletant, at least not the only one, for the bundle formation upon starvation that we observe.

Next, we investigated the influence of OMVs. If OMVs are main depletants, a significant amount of OMVs needs to be released in the medium. Since OMVs are produced by cells, their density is expected to be higher inside bundle clusters, between individual bundles (the region between dashed rectangles in Fig. 2b(ii)). We attempted to inspect this by staining cell membranes and OMVs with FM 4-64 dye, which selectively label hydrophobic cell membranes including OMVs[62,63] (Fig. S5, ESI†). By using the W3110 Δ*fliC intS*::*venus-kan* mutant (''Δ*fliC* (*venus*) mutant'' in Table 1), the cell bodies were also visualized by the fluorescent protein Venus. We observed aggregates of this strain grown in M9(glc. + a.a.) and starved in PBS (''Δ*fliC* (*venus*) starvation'' in Table 2). However, the results in Fig. S5 (ESI†) did not show an appreciable amount of OMVs in the medium, even between individual bundles inside clusters (compare the fluorescence intensities of Venus and FM 4-64). This suggests that OMVs may not be the main depletant that triggers the bundle formation.

As another test, we attempted to induce bundle formation by promoting the secretion of OMVs by antibiotics, instead of starvation. Specifically, we used polymyxin B, known to increase the OMV secretion by several times to tenfold at sub-minimum inhibitory concentrations (sub-MIC).[64] We cultured the Δ*fliC* mutant in growth medium (either M9(glc. + a.a.) or LB) containing sub-MIC polymyxin B for a long time, without starvation, but no bundle clusters were observed at all (''Δ*fliC* growth (PMB)'' and ''Δ*fliC* growth (LB, PMB)'' in Table 2 and Fig. S6a, b, Movies S12, S13, ESI†).

We also attempted to induce the bundle formation by an external addition of a standard depletant, namely xanthan gum, an extracellular polysaccharide produced by *Xanthomonas campestris*,[65] which was used in the literature to introduce depletion attraction between cells.[14] Following ref. 66 and 67, we supplemented xanthan gum to medium, at a concentration sufficient to reproduce the viscosity due to EPS secreted by *E. coli*. We observed aggregates of the Δ*fliC* mutant grown in M9(glc. + a.a.) (''Δ*fliC* growth (xanthan)'' in Table 2). In the early stage while the cell density was low, we observed a few small bundles containing 2–3 cells that seem to be induced by the depletion attraction (see Movie S14, ESI†). However, the small bundles immediately disappeared afterward, and no bundle formation was finally realized in any of the observed wells (Table 2 and Fig. S6c, ESI†). An interpretation of these observations under growth conditions with depletants will be given in the next subsection.

Our series of experiments suggest that, while depletion attraction *via* EPS seems to contribute significantly to the bundle formation upon starvation, the presence of a depletant solely is not sufficient to induce the bundle formation.

2.4 The relevance of growth arrest

Now we consider the possibility that the arrest of cell growth by nutrient starvation may also have a significant effect on the bundle formation. We hypothesize that, even if a sufficient amount of depletants are diffused between cells, perturbations due to cell elongation and division may overcome the depletion attraction and prevent bundle formation.

To test this hypothesis, we attempted to switch back from the non-nutritious to nutritious conditions, after bundle clusters were formed upon starvation. Specifically, we first prepared bundle clusters of the Δ*fliC*, grown in M9(glc. + a.a.) and starved in PBS, as in the Δ*fliC* starvation, then resumed cell growth by switching it back to M9(glc. + a.a.) (''Δ*fliC* starvation-recovery'' in Table 2, Fig. 5a and Movie S15, ESI†). We measured the time evolution of the bundle order parameter in three independent wells, as well as the total size of aggregates in those wells (Fig. 5b). During the starvation stage (PBS), the bundle order parameter $\langle S_B \rangle$ increased, while the growth of the total aggregate size slows down (red dotted line). This decline of the growth rate takes place only gradually, as observed also in our previous work,[32] presumably because cells might continue growing for a while by consuming nutrients remaining inside. Then, after the growth condition was recovered, the aggregate size started to grow again, while the smectic-like bundles were destroyed and the bundle order parameter decreased. We also evaluated the size and number of individual bundle clusters and obtained consistent results; both quantities increased during the starvation and decreased after the nutrient recovery (Fig. 5c). Therefore, not only the depletion attraction induced by EPS but also growth arrest is necessary for the bundle formation we observed. The absence of growth arrest might

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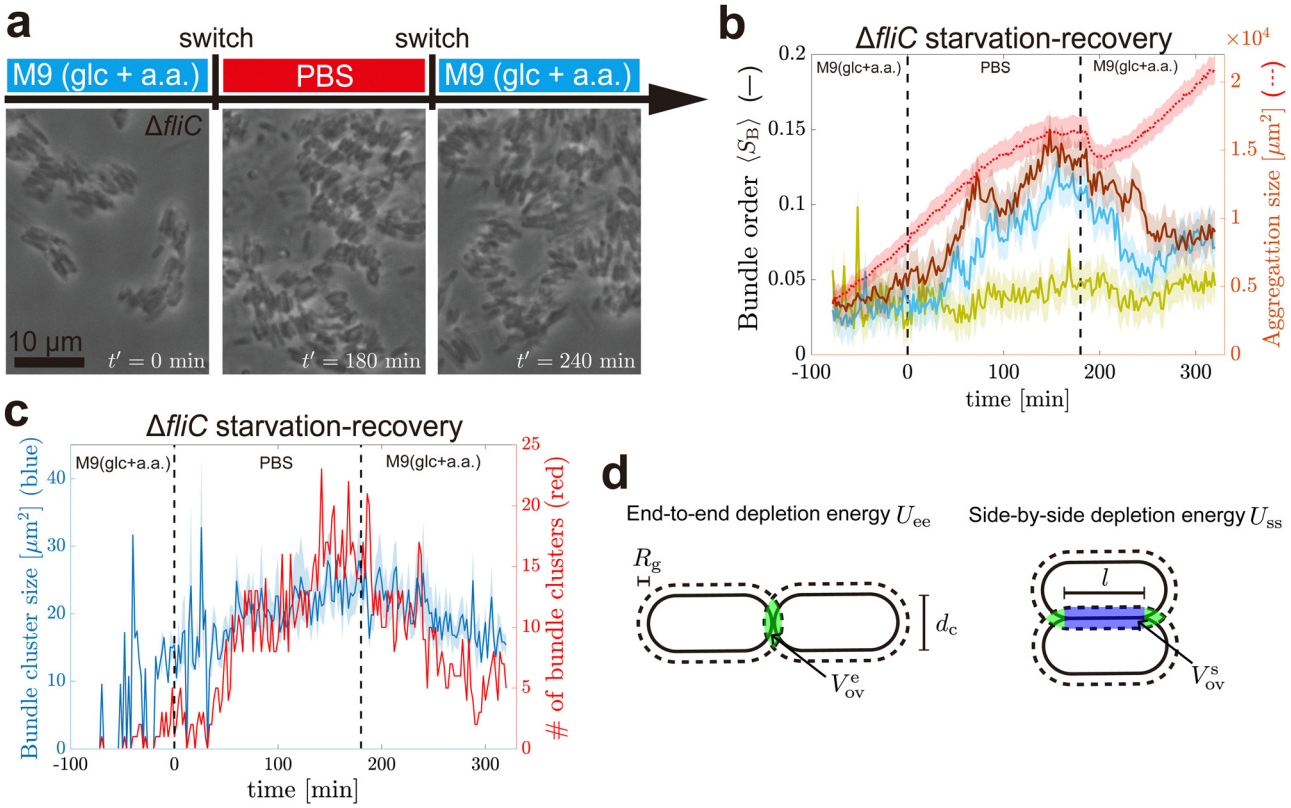


Fig. 5 Investigation of the relevance of cellular growth arrest. (a) Observations of the Δ*fliC* mutant grown in M9(glc + a.a.), starved in PBS, then grown again in M9(glc + a.a.). $t' = 0$ is the time at which the environment was switched to PBS, and the switch back to M9(glc + a.a.) was at $t' = 180$ min. The well diameter ϕ is 150 μm. Bundle clusters were observed in 25 out of 27 wells with various diameters recorded in a single experiment (Δ*fliC* starvation-recovery in Table 2). Those bundles were destroyed after the recovery of cell growth by M9(glc + a.a.) in all wells. The displayed snapshots were taken from the same position. See also Movie S15 (ESI†). (b) The spatially averaged bundle order parameter over the region where aggregates exist, $\langle S_B \rangle$, in the Δ*fliC* starvation-recovery experiment. Each time series represents the data in a separate well (ϕ = 230 μm). The observation shown in (a) and Movie S15 (ESI†) corresponds to the blue curve. The definition of the errors (shades) is the same as that for Fig. 3d. We also plot the aggregation size, summed over the three wells (red dotted line). The displayed error for the aggregation size is the sum of the errors of the individual aggregate sizes. The error of each aggregate size was estimated to be the aggregate perimeter × 5 pixels (~0.85 μm), which we consider to be the range of uncertainty in the edge location of the segmented region. The sudden decrease of the total aggregation size may be due to a weak hydrodynamic disturbance caused by the medium switch. (c) The size and number of bundle clusters for the Δ*fliC* starvation-recovery, measured in all three wells used in (b) at 20-minute intervals. The blue curve and the shaded band indicate the mean and standard error of the cluster sizes, respectively. The red curve represents the number of bundle clusters. (d) Sketch of the end-to-end and side-by-side depletion energies.

be the reason why bundle clusters were not formed in the aforementioned Δ*fliC* growth (PMB), Δ*fliC* growth (LB, PMB), and Δ*fliC* growth (xanthan) experiments, where an increased amount of depletants was present but growth was not arrested.

2.5 Starved cells with externally imposed depletants

Our experiments so far indicate that both the depletion attraction *via* EPS and growth arrest of cells are relevant to the bundle formation we observed. Using this, we carried out controlled experiments to quantitatively determine the concentration of depletant needed to induce the bundle formation. More specifically, we observed bacteria on a glass-bottom dish, starved by PBS in the presence of varying concentrations of externally imposed xanthan gum, while spontaneous EPS secretion was inhibited by BisBAL (see Section 4.6 for details). Then we found that bundle formation appeared at xanthan gum concentrations higher than 0.0075 wt% ("Δ*fliC* starvation-dish (xanthan, BisBAL)" in Table 2 and Fig. S7, ESI†). This underpins our finding that both the depletion attraction and the growth arrest are important for the bundle formation.

Moreover, by using the critical concentration of xanthan gum, we can evaluate the strength of the depletion attraction needed to induce the bundle formation. Following ref. 47, we characterize the depletion attraction between two *E. coli* cells, modeled as spherocylinders of diameter d and length l, by the end-to-end and side-by-side depletion energies (Fig. 5d). The two depletion energies, denoted by U_{ee} and U_{ss}, respectively, are expressed in terms of the excluded volume between two spherical ends and that between two cylindrical sides, V^e_{ov} and V^s_{ov}, respectively, as follows:[68]

$$U_{ee} = -n_{xg} k_B T V^e_{ov}, \tag{2}$$

$$U_{ss} = -n_{xg} k_B T (V^e_{ov} + V^s_{ov}), \tag{3}$$

$$V^e_{ov} = \frac{4\pi}{3} R_d^3 \left(1 - \frac{3}{4}\frac{d}{R_d} + \frac{1}{16}\left(\frac{d}{R_d}\right)^3\right) \tag{4}$$

$$V_{ov}^{s} = lR_d^2\left(2\arccos\frac{d}{2R_d} - \frac{d}{R_d}\sqrt{1-\left(\frac{d}{2R_d}\right)^2}\right), \quad (5)$$

$$R_d = d/2 + R_g, \quad (6)$$

with n_{xg} being the depletant (xanthan gum) number density, R_g its gyration radius, k_B the Boltzmann constant, and T the temperature. Setting d = 1 μm and l = 4 μm as the typical size of *E. coli* and assuming the typical molecular weight ≈ 10^6 g mol^{-1} and gyration radius R_g ≈ 100 nm of xanthan gum,[69] we obtain $U_{ss} \approx -20k_BT$ and $U_{ee} \approx -2k_BT$ for the critical concentration of xanthan gum that we determined, 0.0075 wt%. Interestingly, this value of U_{ee} is compatible with the critical value reported in ref. 47 for cells to form clusters *via* depletion attraction. This quantitatively validates the relevance of the depletion attraction to the bundle formation we observed here.

3 Concluding remarks

In this study, using the EMPS we developed previously, we realized observations of planktonic aggregates of non-motile *E. coli* under time-dependent starving conditions. We discovered that nutrient starvation altered the structure of aggregates, resulting in the appearance of smectic-like bundle clusters, where cells arranged themselves side by side. By utilizing the deep learning method, we successfully quantified the degree of smectic-like order of the bundle clusters. We revealed that EPS produced by cells is involved in the bundle formation, by showing that comprehensive reduction of EPS secretion prevented the bundle formation. We also found that the smectic-like bundles were destroyed by resumption of growth after recovery of nutrients. Our results indicate that both the depletion attraction *via* EPS and growth arrest of cells are relevant to the bundle formation we observed.

The contribution of the depletion effect to the smectic bundle formation can be indeed understood as a result of the minimization of the depleted region, as sketched in Fig. 4a. Previous simulations of mixtures of colloidal rods and spheres[44,48–51] reported the realization of the smectic A phase by such a depletion effect. Therefore, smectic bundles may be formed solely by the depletion effect, in the case of colloidal particles which do not grow and divide. However, further investigation is required for elucidating what specific substances are most relevant to the depletion attraction involved. In this work, we inspected two specific substances, curli and OMVs, but could not determine their relevance conclusively. It is probably difficult to validate the relevance of OMVs by inhibition, because complete inhibition of OMVs seems to be unrealistic for living cells. An alternative approach would be to visualize how OMVs spatially distribute around bundle clusters, by labeling OMVs with, *e.g.*, curvature-sensing peptides.[70] We should also think of the possibility that multiple depletants contribute comparably. The dependence on the amount of depletants may also be interesting. Indeed, a colloid experiment using a mixture of rod-shaped and spherical particles[44] reported the appearance of various stable states, such as nematic and smectic phases, depending on the amount of depletants.

The effect of cell growth preventing bundle formation is also an important conclusion drawn in this work. This finding, together with a recent report of nematic cable formation under growth conditions in polymeric environments,[47] emphasizes the importance of the interplay of cell growth and the depletion effect. This suggests that more diverse aggregation patterns may appear under varying growth rates, in particular when it is comparable to time constants associated with the depletion effect. More generally, cell motility may also affect aggregate morphologies through the interplay with the depletion effect.[21,22] It is an important future task to develop a mechanistic model that can account for our finding quantitatively, in terms of depletion force, cell growth and motility, presumably as an extension of the models studied in ref. 20–22.

In the literature, it is also known that changes in cell surface conditions may affect the formation of similar bundle clusters. For *P. aeruginosa* populations, it was observed that decreasing the hydrophobicity of O-specific antigen on the cell surface resulted in the reduction of the hydrophobic attraction between cells, which facilitated the bundle formation by polymers added in the environment.[45] In the case of *E. coli*, it was reported that cells in the stationary phase develop heterogeneous charge distribution on the outer membranes, reducing the electrostatic repulsions between cells and surfaces.[71] It may also be important to seek for the possible existence of unidentified processes underlying the bundle formation we observed. We also note the apparent similarity to the rouleaux formation of red blood cells.[72]

A more complete characterization of the spatial structures of the observed bundles is also called for. For example, clusters of vertically aligned cells, as shown in Fig. 2c, were not characterized in this work. Investigations of three-dimensional structures may provide further insights on the underlying mechanism behind this bundle formation. It may also be interesting to investigate the possibility of the polar ordering in bundle clusters. The polar ordering of the new and old poles of *E. coli* has been observed in small clusters consisting of a few cells.[36,73] In our bundle clusters, while random polarity is anticipated from the entropic point of view, a polar order might also arise within bundles due to the asymmetric adhesion of the cells.[73] A method to distinguish the new and old poles, such as the ones based on the gene expression level[74] and the ribosome concentration,[75] is interesting in this context.

From more general perspectives, the physiological significance of the bundle formation needs to be investigated in the future. For the case of the nematic ordering of anisotropic cells, previous studies[76,77] proposed its effect on the diffusion efficiency of small molecules, such as nutrients and signal molecules produced by cells, in the intercellular gaps. Specifically, in the case of dense *E. coli* populations, it has been experimentally suggested that the transport efficiency along the director of the nematic order can be enhanced.[77] It may be interesting to consider an analogous effect in the presence of the bundle

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order as we observed in this work. If the bundle formation is beneficial for, *e.g.*, an uptake of nutrients from the external environment under starvation, our finding may be a behavioral adaptation of bacterial populations. Besides, the biological functions of liquid crystalline order – mostly nematic – in cell populations are currently under active investigation,[78,79] including bacterial populations.[80–83] Our finding of the smectic-like order in bacteria may suggest the potential importance of this liquid crystalline state too, calling for further investigations of its relevance from various perspectives.

Finally, our findings may also be relevant to other bacteria, because the elevated EPS secretion and the growth arrest by starvation are not unique to *E. coli*. Further elucidation of the mechanism of the bundle formation and its biological significance is an important future task, which may also contribute to understanding the physical mechanisms underlying biofilm development.

4 Materials and methods

4.1 Strains and culture media

We used the wild-type (WT) *E. coli* strain W3110 and mutant strains W3110 Δ*fliC*, W3110 *flu*::*kan*, W3110 Δ*fliC csgA*::*kan*, and W3110 Δ*fliC insS*::*venus-kan* (see Table 1 for the strain names and their genotypes). We used the Δ*fliC* mutant prepared in ref. 84. By P1 transduction, the genes *flu*::*kan* and *csgA*::*kan* were transferred to the WT and the Δ*fliC* mutant, respectively, from the Keio collection of *E. coli* through National BioResource Project (NBRP) of National Institute of Genetics, Japan. Concerning the strain Δ*fliC* (*venus*) mutant, we used a strain that was previously prepared by a method similar to that in ref. 85. The protein Venus was developed in ref. 40 and provided by the RIKEN BRC through NBRP of the MEXT/AMED, Japan.

Table 2 presents a list of the experiments conducted in this study. The media we mainly used were M9(glc. + a.a.) (glucose 0.2 wt% and MEM Amino Acid solution (M5550, Sigma) 1 wt% in the M9 medium), LB broth (tryptone 1 wt%, sodium chloride 1 wt% and yeast extract 0.5 wt%), and phosphate-buffered saline (PBS).

4.2 Evaluation of the loss of curli by Congo red

We verified the loss of curli in the Δ*fliC csgA*::*kan* mutant as follows. We first inoculated the strain from a glycerol stock into 2 ml LB in a test tube and shook it overnight at 37 °C. We then inoculated 10 μl of the incubated suspension to an LB agar containing 25 μg ml^{-1} of Congo red by an inoculation loop. We then obtained colony images of the Δ*fliC* mutant and the Δ*fliC csgA*::*kan* mutant, confirming the absence of dye fluorescence from the latter mutant, due to the loss of curli (Fig. S4a, ESI†).

4.3 Time-lapse observations

Time-lapse observations were conducted by using the extensive microperfusion system (EMPS) previously developed by us and coworkers.[32] We fabricated 11 μm deep circular wells with varying diameters, ϕ = 110, 150, 230, 330 μm, on a coverslip (Fig. 1a). After introducing cell suspension, these wells were covered by a rigid porous membrane, attached by biotin–streptavidin bonding. The porous membrane is a bilayer of a streptavidin-decorated cellulose membrane (Spectra/Por 7, Repligen, Waltham, MA, molecular weight cut-off: 25 000) and a biotin-decorated PET membrane (obtained from Transwell 3450, Corning, NY, nominal pore size: 0.4 μm). See ref. 32 for more details on the fabrication process of the EMPS.

To prepare bacterial suspensions, we first inoculated the strain from a glycerol stock into 2 ml fresh medium in a test tube and shook it overnight at 37 °C. We used either M9(glc + a.a.) or LB broth, chosen according to the growth medium used in the main experiment (see Table 2). We then transferred 20 μl of the incubated suspension to 2 ml fresh medium (the same medium as that used in the main observation) and cultured it until the optical density (OD) at 600 nm wavelength reached 0.1–0.5. The cell suspension was finally diluted to OD = 0.05 before the inoculation on the substrate.

The device was placed on the microscope stage, in the incubation box maintained at 37 °C (Fig. 1b). We used a Leica DMi8 microscope equipped with a 63× (N.A. 1.30) oil immersion objective lens and operated by Leica LasX. A syringe filled with the nutritious medium was connected to the device, and another syringe filled with non-nutritious buffer was also connected *via* a three-way junction with check valves when observing the starvation response. Each syringe was set to a syringe pump (NE-1000, New Era Pump Systems). To fill the device with the nutritious medium, we first injected the medium stored at 37 °C at a rate of 60 ml h^{-1} for 5 min. During the observation, first the nutritious medium was constantly supplied at a rate of 2 ml h^{-1} (flow speed approximately 0.2 mm s^{-1} above the membrane). When changing the environment, the syringe pump for the nutritious medium was stopped, and the other pump for the non-nutritious buffer was turned on to replace the medium inside the device. The flow rate was set to be 60 ml h^{-1} for the first 5 min, then returned to 2 ml h^{-1}.

Throughout the experiment, the device was always in the microscope incubation box, maintained at 37 °C. Cells were observed by phase-contrast microscopy focusing near the bottom of the wells and recorded at a time interval of 2 min. Images were taken from several wells (stated in the figure captions and Table 2) in each experiment. The fraction of the wells showing bundle clusters presented in Table 2 is based on visual assessment.

4.4 Fluorescence labeling of cellular membranes by FM 4-64

For the observation shown in Fig. 1c and Fig. S5 (ESI†), we first inoculated the strain from a glycerol stock into 2 ml M9(Glc + a.a.) in a test tube and shook it overnight at 37 °C. We transferred 20 μl of the incubated suspension to 2 ml fresh M9(Glc + a.a.) and cultured it until the OD reached 0.1–0.5. The cell suspension was finally diluted to OD = 0.05 before the inoculation on the device substrate.

Preparation of the device and the microscope, and induction of nutrient starvation were conducted similarly to the time-lapse observations. 150 min after the onset of starvation, when

smectic-like bundle clusters appeared, PBS containing 5 μg ml^{-1} of FM 4-64 was introduced. The medium flow was set to be 60 ml h^{-1} for the first 5 min and left for 10 min without flow to stain cellular membranes. We then washed unbound FM 4-64 by supplying PBS without FM 4-64 at a rate of 60 ml h^{-1} for 3 min, stopped the flow, and obtained fluorescence images.

4.5 Image analysis

We employed image analysis to evaluate the smectic-like order of bundle clusters, in terms of the bundle order parameter S_B [eqn (1)] as follows. The entire pipeline of the image analysis is illustrated in Fig. S2 (ESI†).

First we detected the regions where planktonic aggregates existed. We utilized the deep neural network, U-Net,[42] to automate this process. For training and evaluation of the model, we extracted 144 frames from all experiments at approximately constant time intervals. Each image consists of 648 × 483 pixels, with pixel size ≈ 0.18 μm. To prepare the teacher label, the target region was manually annotated by using painting software. We also performed data augmentation through the horizontal and/or vertical flip and separated the data into five groups for cross-validation. After normalizing the image intensity distribution by contrast limited adaptive histogram equalization (CLAHE), U-Net was trained using the binary cross-entropy loss. The segmentation accuracy of the trained model, evaluated by the Dice similarity coefficient (DSC), was 0.577.

We then detected two orientation fields, namely the density correlation director $\boldsymbol{R}$ and the cell orientation $\boldsymbol{N}$, separately. For $\boldsymbol{R}$, we employed the structure tensor method.[41] We calculated the structure tensor $J(\boldsymbol{r})$ at a given pixel $\boldsymbol{r} = (x, y)$ by

$$J(\boldsymbol{r}) = \begin{pmatrix} [\Delta_x I, \Delta_x I]_{\boldsymbol{r}}, & [\Delta_y I, \Delta_x I]_{\boldsymbol{r}} \\ [\Delta_x I, \Delta_y I]_{\boldsymbol{r}}, & [\Delta_y I, \Delta_y I]_{\boldsymbol{r}} \end{pmatrix}, \tag{7}$$

with the image intensity $I(x, y)$, $\Delta_x I \equiv I(x+1, y) - I(x-1, y)$, $\Delta_y I \equiv I(x, y+1) - I(x, y-1)$, and $[g, h]_{\boldsymbol{r}} \equiv \sum_{(x', y') \in \mathrm{ROI}_{\boldsymbol{r}}^{\ell}} g(x', y') h(x', y') f_{\boldsymbol{r}}^{\sigma}(x', y')$. The summation is taken over the region of interest $\mathrm{ROI}_{\boldsymbol{r}}^{\ell}$, which is a square of size $\ell \approx 7.2$ μm (40 pixels) centered at $\boldsymbol{r}$, and $f_{\boldsymbol{r}}^{\sigma}(x', y')$ is the Gaussian kernel defined by $f_{\boldsymbol{r}}^{\sigma}(x', y') \equiv \exp\left[-\frac{(x'-x)^2+(y'-y)^2}{2\sigma^2}\right]$ with $\sigma \approx 1.8$ μm (10 pixels). Then $\boldsymbol{R}(\boldsymbol{r})$ is given by the eigenvector of $J(\boldsymbol{r})$ associated with the smallest eigenvalue.

For the cell orientation $\boldsymbol{N}$, we employed the U-Net again. To obtain the teacher label, we manually annotated the long axis of cells by painting black straight lines (see Fig. S2, ESI†). We then calculated the structure tensor from the painted images to acquire the ground truth of the cell orientation, $\hat{\boldsymbol{N}} = \pm(\cos\hat{\psi}, \sin\hat{\psi})$ with $(-\pi/2 \le \hat{\psi} < \pi/2)$. The U-Net was trained to predict $\boldsymbol{N} = \pm(\cos\psi, \sin\psi)$ for each pixel in the CLAHE processed image, with the following nematic L2 loss function:

$$\ell_{\mathrm{NL2}}(\psi, \hat{\psi}) = \min(|\psi - \hat{\psi}|, |\pi - |\psi - \hat{\psi}||)^2. \tag{8}$$

During training, the loss function was evaluated only in the regions with planktonic aggregates obtained by the segmentation model. The nematic L2 loss was then spatially averaged over all regions with planktonic cells, and the mini batch training with a batch size of 8 was carried out. The angle measurement accuracy of the trained model, evaluated using the mean absolute error of ψ, was ±20.9°. Using the predicted $\boldsymbol{N}$, we also evaluated the nematic order parameter by

$$S_{\mathrm{N}}(\boldsymbol{r}) = \langle \sin 2\psi \rangle_{\mathrm{ROI}_{\boldsymbol{r}}^{\ell}}^2 + \langle \cos 2\psi \rangle_{\mathrm{ROI}_{\boldsymbol{r}}^{\ell}}^2, \tag{9}$$

where $\langle \cdot \rangle_{\mathrm{ROI}_{\boldsymbol{r}}^{\ell}}$ denotes the spatial average within $\mathrm{ROI}_{\boldsymbol{r}}^{\ell}$.

Finally, using all these data, we calculated the bundle order parameter S_B [eqn (1)] for each pixel (see also Fig. S2, ESI†). We obtained the spatially averaged bundle order parameter $\langle S_B \rangle$ over the regions with planktonic aggregates, detected by the segmentation model. The mean absolute error of $\langle S_B \rangle$ per image was ±0.013 and shown as shaded bands in the figures.

We also detected and analyzed bundle clusters, based on the bundle order parameter S_B and the uniformity of the density orientation $\boldsymbol{R}$. We first extracted regions with $S_B > 0.5$, where bundles were formed. Then each connected region was divided into sub-regions characterized by almost uniform $\boldsymbol{R}$. More precisely, we measured the angle distribution of $\boldsymbol{R}$ with a bin width of 3° for each connected region and detected peak angles, which were regarded as the representative angles of the sub-regions. We then split the connected region by assigning each pixel $\boldsymbol{r}$ to the sub-region with the representative angle closest to $\boldsymbol{R}(\boldsymbol{r})$. Sub-regions smaller than 8 μm^2, approximately the total area of two cells, were not considered as bundle clusters and excluded from the analysis.

4.6 Confocal imaging with externally added EPS

To quantitatively characterize the depletion forces acting on bundle formation, we observed aggregate formation at controlled concentrations of externally imposed EPS. To eliminate the effects of self-secreted EPS on bundle formation, we subcultured the overnight-cultured medium of the Δ*fliC* strain into M9(glc. + a.a.) with 2 μM BisBAL, suppressing the EPS production of growing cells, and cultured it until the OD at 600 nm reached 0.1–0.5. Then we replaced the medium with PBS containing 2 μM BisBAL, cell membrane-staining FM 4-64 (5 μg ml^{-1}) for cell imaging, and different concentrations of xanthan gum (0–0.03 wt%) as externally added EPS, by centrifuging the cultured medium and suspending the pellet in 200 μL of this starvation medium to induce growth arrest. This suspension filled the base of a glass-bottom dish (with a 0.16–0.19 μm thick cover glass and a well with a diameter of 14 mm and an approximate height of 1 mm, D11531H, Matsunami Glass) and was covered with an overlying PDMS slab, and its plastic lid was sealed with Parafilm (Amcor). We followed ref. 47 for this method using a glass-bottom dish. The suspension was then incubated at 37 °C either in a stage-top incubator on a microscope for time-lapse observation or in an incubator for fixed time-point observation. For time-lapse observation, the brightfield and fluorescence images of cells in the dish were

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taken using a Leica SP8 confocal microscope with a 100× oil-immersion objective (HC PL APO C52, N.A. 1.40, # 11506372, Leica) under Leica LasX operation with 400 Hz scan speed, 30-min time intervals for 18 h, 1 μm *z*-stack intervals for 5 μm image height, and 1 Airy unit for the confocal pinhole. The image size was 512 × 512 pixels with pixel size ≈ 0.23 μm. The FM 4-64 fluorescence images were taken using a 552 nm laser (intensity: 1% or 2%) for excitation and a 570–650 nm detection range for emission with a gain of 100. The brightfield images were taken simultaneously with the fluorescence images with a gain of 350. We imaged three positions of each dish and repeated the same experiment using three different dishes for each xanthan gum concentration. We picked up one *z*-plane of the stack images to show the results in the figures. For fixed time-point observation, cells in the dish were imaged as described above without any time interval, 18 h after the cultivation was started.

Author contributions

Takuro Shimaya: conceptualization, software, formal analysis, funding acquisition, investigation, methodology, writing – original draft. Fumiaki Yokoyama: investigation, methodology, funding acquisition, writing – original draft. Kazumasa A. Takeuchi: conceptualization, funding acquisition, project administration, writing – review & editing.

Data availability

The code and data that support the findings of this study are available at Zenodo. DOI: **https://doi.org/10.5281/zenodo.14984846**.

Conflicts of interest

There are no conflicts to declare.

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